

Indugu Duryodhan Rao

Lead Data Engineer | 15+ years IT experience

Duryodhan.y2k@gmail.com | t: +12102786467 | www.linkedin.com/in/duryodhanraoindugu | <https://iduryodhanrao.github.io>

TECHNICAL SKILLS

- **Programming:** Python (PySpark, Scikit-learn, Tensorflow, PyTorch), SQL, Unix, Java, SAS
- **Data Platforms:** Snowflake, BigQuery, Hadoop, PySpark, Kafka, NoSQL, DB2, Oracle, Netezza
- **ETL & Modeling:** DBT, Datastage, Apache Airflow, Erwin, Control-M
- **ML/AI:** Natural Language processing, Predictive analysis, Statistical analysis, Gemini, Cortex
- **DevOps & Cloud:** AWS, GCP, Terraform, Docker, Kubernetes
- **Data Visualization:** Tableau, SAP Business Objects, Cognos

PROFESSIONAL EXPERIENCE

USAA, San Antonio, TX

June 2018 – Present

Provider of Insurance, banking, investment and retirement solutions exclusively to members of military, veterans and their families.

Senior/Lead Data Engineer

- **Architected and led the migration of 12 legacy platforms** (Hadoop, DB2, Oracle, Essbase, Datastage, Informatica) to a cloud-native ecosystem (Snowflake, AWS, DBT), enhancing system observability and scalability while reducing infrastructure and maintenance costs by **37%** and ETL runtime by **60%**.
- **Defined and executed historical data movement strategy** for hundreds of terabytes, implementing reconciliation and validation frameworks to ensure data integrity and zero loss across mission-critical systems.
- **Designed a reusable ETL conversion framework**, standardizing transformation logic and reducing development effort by **70%** across cross-functional engineering teams.
- **Built predictive ML pipelines** to support marketing product acquisition, delivering actionable insights that drove a **15% increase in lead conversion efficiency**.
- **Developed LLM-powered diagnostic agents** using Snowflake Cortex AI to surface member friction points, directly supporting CTO-led operational strategy and boosting CX metrics.
- **Engineered an enterprise-wide sales funnel** by integrating data from six business units and multiple channel sources, enabling spend optimization, attribution modeling, and ROI measurement.
- **Built SLO-driven Snowflake pipelines and Tableau dashboards** supporting 200+ analysts, improving trust in data accuracy, availability, and business-critical decision-making.
- **Led a team of five Tableau administrators** for the ESM site, streamlining dashboard deployments, optimizing rendering performance, and implementing asset governance—resulting in **50% faster deployments** and **zero downtime**.
- **Provided architectural guidance and mentorship** on Snowflake and DBT best practices, creating demo documentation and reusable templates that accelerated onboarding and standardized deployment across the department.
- **Partnered cross-functionally with business and technology teams**, translating complex technical solutions into executive-level insights and aligning platform roadmaps with strategic priorities.

HHS North Dakota—Deloitte Consulting, Bismark, ND

June 2016 - May 2018

Delivers Integrated health, economic and social support services statewide

Senior ETL consultant

- Implemented an Integrated Eligibility ETL and analytics platform, modernizing legacy infrastructure across CCAP, SNAP, Medicaid, and TANF programs – resulting in 60% faster eligibility processing and 30% reducing in data reconciliation errors.
- Streamlined cross-agency data integration and partnered with SMEs to architect scalable pipelines, enabling real-time reporting and multi-program visibility across health and human services.

- Led user training and iterative feedback cycles, achieving 95% adoption rate, while ensuring compliance with federal reporting standards.

USAA, San Antonio, TX

Jan 2010 - May 2016

Data Engineer

- Implemented campaign data systems enabling precision targeting across member segments and geographies, driving measurable lift of 28% in engagement and 15% increase in conversion rate.
- Automated outbound campaign pipelines and response analytics, reducing manual effort by 40% and accelerating campaign deployment cycles by 3x.
- Led multi-phase platform migrations (Ephiphany → Unica → Adobe Campaign), enabling advanced personalization and improving marketing ROI.
- Engineered a centralized Application Risk Management data mart integrating life cycle data across network, device, and software assets, supporting \$10M+ in infrastructure planning and upgrade forecasting
- Designed health scoring framework and Tableau dashboards to forecast licensing and infrastructure needs, enabling zero unplanned downtime and 100% compliance with quarterly audit requirements.
- Delivered executive-level reporting and insights that informed strategic infrastructure decisions and licensing negotiations, directly supporting CTO-led transformation initiatives and multi-quarter budget alignment.

CNA Financial Corporation – Syntel, Pune, India

Oct 2006 - Jan 2010

Provider of commercial insurance products including liability, auto, workers' compensation.

Software Engineer

- Built a workers' compensation reporting system from the ground up on mainframe architecture, supporting \$100M+ in annual claims volume and achieving 99.9% reporting accuracy
- Delivered end-to-end solution design and implementation, streamlining claims adjudication workflows and reducing processing time by 30%, resulting in \$500k+ in annual operational savings.

Education

Master of Science, Data Science, 2021

MBA, Information Technology management, 2024

Bachelors of Engineering, Electronics, 2006

Certification

Cloud & Devops: AWS DevOps Professional, Snowflake SnowPro, Terraform

ETL, Analytics & Data Engineering: DBT Developer, SAS Base Programmer, SAS statistical, Datastage

Programming and Process Excellence: Python Pro, Six Sigma Green Belt

Personal Projects – Github portfolio highlights

- **LLM & ML Workflows:** Built diagnostic and predictive pipelines using Jupyter Notebooks (ml-tests, google-gemini-agents, MLOPS). Includes model evaluation, prompt engineering, and cloud ML integration.
- **Live Stock Market Dashboard:** Python utility for real-time financial data aggregation and visualization (realtime_stockmarketsummary). Highlights API handling and data streaming.
- **Retail Revenue Analytics:** SAS dashboard for SKU-level revenue tracking and regional trends (retail_revenue). Applied statistical methods and visualizations.
- **Fleet Service Scoring Engine:** Python-based ranking system for vehicle service providers (car_rankings). Focused on algorithm design and user-centric reporting.
- **Multi-Source Utility Suite:** Unified Python interface integrating weather, financial, and news data streams to simulate real-time decision support—enabling agentic AI prototyping and cross-domain analytics acceleration